

TM10011: Machine Learning

Project guidelines

January 30, 2026

1 Introduction

This guide describes how to conduct and report on the final research project for the TM10011 Machine learning course. In this project, you will use one of the provided medical datasets to train and evaluate different classification methods.

2 Goal of the project

The goal of the project is twofold. The first goal is to develop a machine learning method from the dataset that you are given. The second goal is to evaluate this model with the same dataset, in a way that you can quantify the performance as reliably as possible. Since both goals are achieved with the same dataset, you need to design a proper experimental method to achieve both. This is a common issue in machine learning research.

In the design of the machine learning method, containing preprocessing, possibly some feature extraction and a classifier, we expect you to motivate your design choices by either experiments or theoretical knowledge, or both. Of course, it is impossible to try every variation of every hyperparameter. Focus, therefore, on a number of relevant choices and evaluate their influence on performance. In the report, describe the experiments you did, show the results and interpret them. Keep in mind that the final grade does not depend on the performance of the model, or the number of methods presented, but rather on the quality of the experimental work and the insight shown in the report and the oral exam.

3 Deliverables

3.1 Written Report

You will have to hand in a written report like a paper, shortly introducing the problem, and describing your methods, results, and a discussion. The report should

have no more than 5 pages, excluding references and appendices, and should contain the following:

1. A title and the names of your group members. Just put this on the top of the page, you don't need to make a separate front page.
2. The link to your github repository.
3. An introduction concerning the (clinical) problem to be solved and the potential role of machine learning in this aspect.
4. A description of the methods applied, with proper motivation, containing for example:
 - A description of the dataset
 - Preprocessing of the data
 - Classifier(s)
 - Experimental and evaluation setup

Your methods should be sufficiently detailed for another person to replicate your study.

5. A results section, providing a comprehensive overview of the results. When possible, try to illustrate your results with plots or tables.
6. A discussion, answering the following questions
 - What performance would this method reach on similar unseen data?
 - What are the limitations of the study and what could be done to improve it?
 - Can this model be applied in the clinic? If not, what further steps could be taken to make it applicable in a clinical setting?

We do not expect you to perform a statistical analysis of the results, but we do expect you to perform the necessary experiments to reach a conclusion on the most promising method and expected performance. If possible, try to relate to your knowledge of the methods used. If possible, make concrete suggestions for further research.

3.2 Code

At time of the deadline, the code has to be uploaded to your groups Github repository and tagged with the tag *final*. The link to your Github repository should be in your report.

3.3 Reflection

To be uploaded separately and individually: a reflection on the collaboration within

the team with the planning, communication strategy, and the division of roles and tasks. Each team member describes his/her individual contribution to both end products and his/her opinion on the process of this teamwork, the positive elements and the negative aspects of the combination of this team with this assignment. The reflection should be around 250 words and must be handed in separately and individually; hence, this does not count for the report's page limit.

4 Grading and assessment

When grading the report, we will take the following aspects into account:

- Is proper motivation supplied for the methodological choices?
- Was a proper experimental setup applied and described in the report?
- Are the results interpreted well, with respect to the different methods?
- Is the clinical relevance of the results discussed?
- Are the conclusions supported by quantitative results and figures?

Note that although a proper analysis and smart choice of methodology will lead to a classifier with high performance, a better performance does not automatically lead to a higher grade.

In the oral examination, questions about the code, the setup, the results, figures etc will be asked to all team members individually. In this project we encourage shared responsibility: the team grade aligns with the lowest individual performance.

5 Deadlines

The deadline will be announced on BrightSpace.

6 Datasets

All datasets contain quantitative medical image features, on which you are going to apply machine learning to find a relation between these and clinical outcome. This field of research is also known as radiomics.

For each dataset, you will have access to a set of these quantitative imaging features, on which you have to perform a binary classification problem.

6.1 Distinguishing lipoma from well-differentiated liposarcoma [dataset 1: worclipo]

Soft tissue tumors (STT) are a rare and complex group of tumors with a broad range of differentiation. According to the WHO guidelines, there exist more than 100 different

STT subtypes [6]. Lipomatous tumours are the most commonly observed soft tissue tumours, mostly owing to the high incidence of benign lipomas. Also within the malignant spectrum of soft tissue tumours (soft tissue sarcomas), liposarcoma is among the most frequently observed subtype.

Whereas lipomas do not necessarily need to be excised, patients with WDLPS are generally considered candidates for surgery. Hence, an accurate diagnosis is required to select the right treatment and follow-up. Currently, the standard way to differentiate lipoma from WDLPS is through a biopsy, which is tested for MDM2 amplification. Taking a biopsy is an invasive and painful procedure for the patient, and is associated with risks, depending on tumour location, and potential sampling error.

The aim of this study is therefore to distinguish lipoma from well-differentiated liposarcoma based on features extracted from T1-weighted MRI. A good performance on this dataset would be above 70% mean accuracy. However, you might want to consider a different performance metric to evaluate the results. The MRI data for this study was obtained from the WORC database [3].

6.2 Distinguishing benign from malignant primary solid liver tumours [dataset 2: worcliver]

Liver cancer is the seventh most commonly diagnosed cancer and the third most common cause of cancer death worldwide, with approximately 906,000 estimated new cases and 830,000 deaths in 2020 [5]. One of the most important tasks in routine clinical practice is to distinguish malignant from benign primary solid liver lesions, as this influences treatment planning. A first assessment is usually made by radiologists based on MRI, but this is often a challenging task, especially in non-cirrhotic livers. The challenges arise due to the wide variety of phenotypes and lack of a clear assessment consensus.

Diagnosis on MRI is commonly based on a variety of sequences, e.g., T2-weighted MRI, T1-weighted MRI with or without (liver-specific) contrast agents, and diffusion weighted imaging. While guidelines exist, the radiologist's interpretation of T2-weighted MRI interpretation remains challenging, qualitative, and observer-dependent. T2-weighted MRI is frequently used and widely available in the assessment of liver cancer, straightforward to perform, and shows less heterogeneity and is less invasive compared to sequences using contrast agents. Hence, a T2-weighted MRI based radiomics model would be widely applicable, scalable, and generalizable.

The aim of this study is therefore to distinguish benign from malignant primary solid liver tumours in non-cirrhotic livers using features extracted from T2-weighted MRI. A good performance on this dataset would be above 70% mean accuracy. However, you might want to consider a different performance metric to evaluate the results. The MRI data for this study was obtained from the WORC database [3].

6.3 Distinguishing gastrointestinal stromal tumors (GISTs) from similar non-GIST sarcoma [dataset 3: worcgist]

Soft tissue tumors (STT) are a rare and complex group of tumors with a broad range of differentiation. According to the WHO guidelines, there exist more than 100 different STT subtypes [6]. One of these are gastrointestinal stromal tumors (GISTs), rare mesenchmal tumors of the gastrointestinal tract. Differentiating GISTs from other intra-abdominal tumors (non-GISTs), i.e., the differential diagnosis, is highly important for early diagnosis and treatment planning. However, due to the rarity of GISTs, establishing a correct diagnosis is challenging. The aim of this study is therefore to distinguish GISTs from non-GISTs based on features extracted from T1-weighted MRI. A good performance on this dataset would be above 60% mean accuracy. However, you might want to consider a different performance metric to evaluate the results. The MRI data for this study was obtained from the WORC database [3].

6.4 Predicting tumor stage in head and neck cancer [dataset 4: hn]

The annual incidence of head and neck (H&N) cancer is around 550.000 cases worldwide per year, with a mortality rate of 300.000 per year. This cancer comes in various forms, and therefore a wide spread in the prognosis and survival of H&N tumors. One of the most important clinical biomarkers is the tumor stage or so called T-stage. This biomarker is used to judge the malignancy of the tumor, and is thus closely related to survival. While T-stage is highly related with volume, the molecular profile of the tumor does also play a role in the malignancy. However, quantification of the molecular profile can only be obtained after a biopsy or resection. While imaging, mostly computed tomography (CT) may be used to quantify T-stage, manual rating is observer dependent, subjective, and challenging.

The aim of this study is therefore to predict the T-stage (high/low) in patients with H&N cancer based on features extracted from CT. A good performance on this dataset would be above 70% mean accuracy. However, you might want to consider a different performance metric to evaluate the results.

6.5 Diagnosing heart disease in 12-lead ECG data [dataset 5: ecg]

The electrocardiograms (ECG) is an important diagnostic tool in cardiovascular diseases. The aim of this study is to automatically find abnormalities on 12-lead ECG. The problem is formulated as a binary prediction between normal and abnormal ECG's, where the abnormalities could be a 1st degree AV block, right bundle branch block, left bundle branch block, sinus bradycardia, atrial fibrillation or sinus tachycardia [1]. The majority of samples contain no abnormalities. Frequency features were extracted from the 12 leads by means of a Fourier transform. From the resulting 9000 features for 827 patients, it is possible to achieve a mean accuracy above 85%. However, you might want to consider a different performance metric to evaluate the results.

References

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